

$$11) \ y = \frac{2}{3}x^4 + 5x - x^{-3}$$

$$12) \ y = -\frac{1}{2}x^4 + 3x^{\frac{5}{3}} + 2x$$

Differentiate each function with respect to the given variable.

$$13) \ y = -3r^5 - 5r^2$$

$$14) \ f(s) = -\frac{3}{s^2} - \frac{4}{s^4}$$

$$15) \ f(x) = \frac{2}{3}x^{\frac{3}{2}} - \frac{3}{4}x^{\frac{3}{5}}$$

$$16) \ h(s) = \sqrt{2} \cdot \sqrt[3]{s} + \sqrt{2} \cdot \sqrt[5]{s}$$

Differentiate each function with respect to x . Problems may contain constants a , b , and c .

$$17) \ y = 5c$$

$$18) \ y = 4ax^{3a} - bx^{3c}$$